

Project_Submission_Point

Rubric Details

- **Data Cleansing, Visualization, and Understanding**

30% of total mark

Outstanding

Outstanding and comprehensive data preparation. Missing values, inconsistencies, and irrelevant features are handled rigorously with strong justification. Excellent exploratory analysis supported by clear and insightful visualisations. Demonstrates deep understanding of the dataset, with meaningful interpretation of patterns and relationships that directly inform modelling decisions.

30

Excellent++

Highly thorough data cleansing and preprocessing with minimal errors. Strong and well-justified handling of missing data and feature selection. Visualisations are clear, relevant, and insightful. Demonstrates a very strong understanding of the dataset, with effective interpretation of key patterns.

27.6

Excellent+

Comprehensive data cleaning with appropriate handling of inconsistencies and missing values. Good justification of preprocessing decisions. Visualisations are clear and support analysis well. Demonstrates strong understanding of the dataset, with relevant insights identified.

26.4

Excellent

Very good data preparation with most issues addressed correctly. Some justification provided, though not always fully developed. Visualisations are appropriate and mostly clear. Demonstrates good understanding of the dataset and key relationships.

24.6

Strong First

Good coverage of data cleansing techniques. Most inconsistencies and missing values are handled appropriately. Visualisations are clear and support the analysis. Demonstrates solid understanding, though insights or justification could be more developed.

23.4

Clear First

Adequate and correct data cleaning with most key steps completed. Some justification or depth of analysis is limited. Visualisations are present and generally appropriate. Demonstrates a clear understanding of the dataset.

21.6

Upper Second (Strong)

Good attempt at data cleaning with appropriate techniques applied. Some missing values or inconsistencies remain insufficiently handled. Visualisations are relevant but may lack depth or clarity. Demonstrates reasonable understanding with some useful insights.

20.4

Upper Second (Clear)

Basic data cleansing applied with some important steps missing or weakly justified. Handling of missing data or feature selection is partially effective. Visualisations are present but limited in insight. Understanding of the dataset is evident but not fully developed.

18.6

Lower Second (Strong)

Limited data cleaning with several issues not fully addressed. Some attempt to handle missing values, but methods may be inconsistent or weakly justified. Visualisations are basic and offer limited insight. Understanding of the dataset is partial.

17.4

Lower Second (Clear)

Minimal data preparation with several major issues remaining. Missing data handling is weak or unclear. Visualisations are simplistic or poorly chosen. Limited understanding of the dataset and its structure.

15.6

Strong Third

Limited and inconsistent data cleaning. Important preprocessing steps are missing or incorrectly applied. Visualisations are unclear or poorly interpreted. Weak understanding of the dataset.

14.4

Clear Third

Very minimal data cleaning with significant errors or omissions. Missing values and inconsistencies are largely unaddressed. Visualisations are ineffective or absent. Very limited understanding of the dataset.

12.6

Marginal Fail

Inadequate data preparation. Major issues with missing values, inconsistencies, and feature handling remain unresolved. Visualisations are absent or incorrect. Demonstrates little understanding of the dataset.

10.5

Very Poor

Very poor attempt at data cleaning and exploration. Work is largely incomplete or incorrect. No meaningful visualisations or analysis provided.

4.5

No submission

No relevant work submitted for this criterion.

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- **Model Building and Evaluation**

20% of total mark

Outstanding

Outstanding model development and evaluation. Multiple well-justified models are implemented with strong feature selection and preprocessing. Appropriate validation strategy is rigorously applied. Class imbalance is effectively addressed. Evaluation uses a

comprehensive range of metrics with deep, insightful interpretation and clear comparison of model performance.

20

Excellent++

Highly effective model development with strong justification of model choices and features. Cross-validation is correctly applied and class imbalance is handled appropriately. Multiple evaluation metrics are used with clear and insightful interpretation. Strong comparison between models.

18.4

Excellent+

Well-developed models with appropriate feature selection and justification. Validation approach is suitable and correctly implemented. Class imbalance is addressed. Good use of multiple evaluation metrics with clear interpretation and comparison.

17.6

Excellent

Very good model implementation with reasonable justification. Validation is applied but may lack depth. Class imbalance handling is present but could be improved. Evaluation uses relevant metrics, though interpretation may not be fully developed.

16.4

Strong First

Good model development with appropriate choice of models. Basic justification provided. Validation and evaluation are applied correctly. Some attempt to handle class imbalance. Interpretation is clear but could be deeper.

15.6

Clear First

Adequate model development with correct implementation. Some justification of choices is provided. Basic validation and evaluation are performed. Limited but appropriate use of metrics. Some gaps in interpretation or comparison.

14.4

Upper Second (Strong)

Good attempt at model building with reasonable choice of algorithms. Limited justification of feature selection or model choice. Validation is present but basic. Evaluation uses some metrics but lacks depth. Model comparison is present but not fully developed.

13.6

Upper Second (Clear)

Models are implemented but with weak justification. Validation approach is basic or partially correct. Evaluation relies on limited metrics (often accuracy-focused). Limited comparison between models. Understanding is evident but not well developed.

12.4

Lower Second (Strong)

Basic model implementation with minimal justification. Validation is weak or inconsistently applied. Evaluation is limited and lacks appropriate metrics. Model comparison is superficial. Some understanding is shown but key elements are missing.

11.6

Lower Second (Clear)

Simple model implementation with little or no justification. Validation is poorly applied or missing. Evaluation is minimal and relies on inappropriate metrics. Very limited comparison or interpretation.

10.4

Strong Third

Weak model development with incorrect or inappropriate choices. Validation is absent or incorrectly applied. Evaluation is poor with little use of appropriate metrics. Limited understanding of modelling process.

9.6

Clear Third

Very weak implementation of models. Major issues with validation and evaluation. Metrics are missing or incorrectly used. Little evidence of understanding.

8.4

Marginal Fail

Inadequate model development. Models are incorrectly implemented or incomplete. No proper validation or evaluation. Very limited understanding demonstrated.

7

Very Poor

Very poor or incomplete attempt at model building. No meaningful evaluation or justification provided.

3

No submission

No relevant work submitted for this criterion.

0

- **Improved Model, Clustering, and Evaluation**

20% of total mark

Outstanding

Outstanding implementation of distributed analysis using PySpark, demonstrating a clear understanding of large-scale data processing. Clustering is rigorously applied with well-justified selection of cluster number (e.g., elbow/silhouette methods). Clusters are clearly visualised and meaningfully interpreted. Cluster-based classification models are effectively developed and evaluated, with strong comparison against the global model. Insightful discussion demonstrates when and why clustering improves predictive performance.

20

Excellent++

Highly effective use of PySpark with clear demonstration of distributed processing benefits. Clustering is well implemented with appropriate justification of cluster selection. Visualisations are clear and informative. Cluster-based models are developed and evaluated with strong comparison to the global model. Good level of insight in interpretation.

18.4

Excellent+

Strong implementation of clustering and distributed processing. Cluster number is reasonably justified. Visualisations are appropriate and support interpretation. Cluster-based classification is implemented and compared with the global model. Good discussion of results.

17.6

Excellent

Very good clustering and PySpark usage, though justification or depth of analysis may be limited. Cluster-based classification is implemented with some comparison to the global model. Visualisations are present but could be clearer. Interpretation is generally sound.

16.4

Strong First

Good clustering implementation with basic justification of cluster number. PySpark is used but may not be fully leveraged. Cluster-based models are implemented with some comparison to the global model. Interpretation is clear but lacks depth.

15.6

Clear First

Adequate clustering and PySpark implementation. Limited justification of cluster selection. Cluster-based classification is present but basic. Some comparison with global model provided. Interpretation is acceptable but limited.

14.4

Upper Second (Strong)

Reasonable clustering implementation with limited justification of cluster number. PySpark is used at a basic level. Cluster-based models are implemented but comparison with the global model is weak or incomplete. Visualisation and interpretation are present but limited.

13.6

Upper Second (Clear)

Basic clustering applied with little justification. PySpark usage is minimal or not clearly demonstrated. Cluster-based classification is weak or partially implemented. Limited comparison and interpretation of results.

12.4

Lower Second (Strong)

Limited clustering implementation with weak justification of cluster number. PySpark usage is superficial. Cluster-based models are poorly implemented or incomplete. Minimal comparison with global model. Interpretation is weak.

11.6

Lower Second (Clear)

Very basic or incorrect clustering approach. Little or no justification provided. PySpark is barely used or misapplied. Cluster-based classification is missing or ineffective. Very limited interpretation.

10.4

Strong Third

Weak clustering implementation with incorrect or poorly applied methods. No meaningful justification of cluster number. PySpark usage is incorrect or absent. Cluster-based classification is not properly developed. Very limited understanding.

9.6

Clear Third

Very poor clustering and distributed analysis. Major errors in implementation. No meaningful comparison or interpretation. Little evidence of understanding.

8.4

Marginal Fail

Inadequate or incomplete clustering and model improvement. PySpark is not used correctly or at all. No meaningful evaluation or comparison provided.

7

Very Poor

Very poor or incomplete attempt. No valid clustering or distributed analysis performed.

3

No submission

No relevant work submitted for this criterion.

0

- **Coding Style, use of appropriate Data Structures and GitLab.**

10% of total mark

Outstanding

Exceptionally well-structured and professional code. Clear modular design with well-defined functions and meaningful naming throughout. Efficient and appropriate use of data structures. Code is clean, readable, and fully reproducible. GitLab is used effectively with multiple meaningful commits reflecting different stages of development. Demonstrates excellent programming practice.

10

Excellent++

Highly organised and readable code with strong use of functions and appropriate data structures. Naming is clear and consistent. Code is reproducible and well-documented where necessary. GitLab is used effectively with clear and meaningful commits.

9.2

Excellent+

Well-structured code with good modularisation and appropriate use of functions. Naming is generally clear. Code is readable and mostly reproducible. GitLab usage is evident with meaningful commits.

8.8

Excellent

Very good code structure with some modularisation. Use of functions is present but could be improved. Naming and readability are generally good. GitLab is used but commit quality may vary.

8.2

Strong First

Good structure, but could be more modular. Some appropriate data structures are used, but minor inefficiencies exist. GitLab is used, but commit history is minimal or lacks clarity.

7.8

Clear First

Adequate code structure with basic use of functions. Some issues with organisation or naming. Code runs correctly but lacks refinement. GitLab is used but commits may be limited or lack clarity.

7.2

Upper Second (Strong)

Code is functional but lacks clear structure in places. Limited modularisation and inconsistent use of functions. Naming may be unclear at times. GitLab usage is basic with few meaningful commits.

6.8

Upper Second (Clear)

Basic code structure with minimal use of functions. Readability is inconsistent. Limited use of appropriate data structures. GitLab usage is weak or poorly evidenced.

6.2

Lower Second (Strong)

Code is poorly structured with little modularisation. Functions are rarely used. Naming is unclear and readability is limited. GitLab usage is minimal.

5.8

Lower Second (Clear)

Very basic or disorganised code. Minimal use of functions or appropriate structures. Code may run but is difficult to follow. GitLab usage is absent or ineffective.

5.2

Strong Third

Weak and poorly organised code. Significant issues with readability and structure. Little or no use of functions. Code may contain errors or inconsistencies.

4.8

Clear Third

Very poor code structure. Code is difficult to understand and may not run correctly. No meaningful use of functions or appropriate data structures.

4.2

Marginal Fail

Inadequate code with major structural issues. Code does not run correctly or is largely incomplete. No evidence of good programming practice.

3.5

Very Poor

Very poor or incomplete code. No meaningful implementation.

1.5

No submission

No relevant code submitted.

0

- **Report Writing and Interpretation**

20% of total mark

Outstanding

Exceptionally well-structured and professionally written report. Presents a clear and logical narrative from data exploration through to modelling and evaluation. All decisions are rigorously justified, with deep and insightful interpretation of results. Strong integration of visualisations into the discussion. Demonstrates critical thinking, clear comparison of models, and excellent understanding of the problem context.

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Excellent++

Highly coherent and well-organised report with strong academic writing. Clear justification of all key decisions and strong interpretation of results. Visualisations are well integrated and support the discussion effectively. Demonstrates excellent understanding and critical analysis.

18.4

Excellent+

Well-structured report with clear flow and good use of academic writing. Decisions are generally well justified. Interpretation of results is clear and meaningful. Visualisations support the analysis effectively. Good comparison and discussion of findings.

17.6

Excellent

Very good report with logical structure and clear writing. Some justification of decisions is provided, though not always fully developed. Interpretation is generally sound. Visualisations are relevant but may not always be fully integrated into discussion.

16.4

Strong First

Good report with clear structure and readable writing. Key steps are explained, though justification may be limited in places. Interpretation is present but could be more insightful. Some comparison of results is provided.

15.6

Clear First

Adequate and well-organised report. Basic explanation of steps and results. Some interpretation is provided, but lacks depth. Limited justification of decisions. Visualisations are present but not always well discussed.

14.4

Upper Second (Strong)

Generally clear report with reasonable structure. Basic explanation of methods and results. Interpretation is present but limited in depth. Justification of decisions is weak or inconsistent. Some gaps in linking results to conclusions.

13.6

Upper Second (Clear)

Report is understandable but lacks strong structure or flow. Explanation of methods is basic. Interpretation is limited and often descriptive rather than analytical. Minimal justification of decisions. Weak integration of visualisations.

12.4

Lower Second (Strong)

Report covers key sections but lacks clarity and depth. Limited explanation of methods and weak interpretation of results. Justification is minimal. Visualisations are included but not well explained.

11.6

Lower Second (Clear)

Basic and incomplete report. Poor explanation of methods and results. Interpretation is weak or missing. Little or no justification. Visualisations are poorly used or unclear.

10.4

Strong Third

Weak report with poor structure and limited clarity. Minimal explanation of methods. Interpretation is largely absent or incorrect. Very limited understanding demonstrated.

9.6

Clear Third

Very poor report with major gaps in content and structure. Little or no meaningful interpretation. Work is largely descriptive or incomplete.

8.4

Marginal Fail

Inadequate report. Major sections are missing or poorly developed. No meaningful interpretation or justification. Very limited understanding demonstrated.

7

Very Poor

Very poor or incomplete report. No coherent structure or meaningful content.

3

No submission

No report submitted.

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